

Loops in JS

Loops are used to execute a piece of code again & again

for Loop

```
for (let i = 1; i <= 5; i++) {  
    console.log("apna college");  
}
```

Loops in JS

Infinite Loop : A Loop that never ends

Apna College

Loops in JS

while Loop

```
while (condition) {  
    // do some work  
}
```

Loops in JS

do-while Loop

```
do {  
    // do some work  
} while (condition);
```

Loops in JS

for-of Loop

```
for (let val of strVar) {
```

```
  //do some work
```

```
}
```

new variable

string variable

used to put for loop for sepcial datatypes like strings and arryas

Loops in JS

for-in Loop

```
for (let key in objVar) {  
    //do some work  
}
```

this loop is used for objects,, unlike
prev one for of loop

used to return keys name of a
object

not keys value but keys name

Let's Practice

Qs1. Print all even numbers from 0 to 100.

```
for(let i=0;i<=100;i++){
```

```
  if(i%2==0){  
    console.log(i);
```

```
  }  
}
```

Let's Practice

Qs2.

Create a game where you start with any random game number. Ask the user to keep guessing the game number until the user enters correct value.

```
let num=28;  
let guess=prompt("enter number");  
  
while(guess!=num){  
  guess=prompt("enter a number");  
}  
console.log("correct u won");
```

do while loop is wrong
coz condition starts from the
second case what if user
write correct ans in first try
then it will also run , so while
loop to be used

here it doesnt ends with \0 like in c

Strings in JS

String is a sequence of characters used to represent text

Create String

```
let str = "Apna College";
```

you're already using it from ch1 like `let name="prince";`
their identification is that they're always inside double quotes

String Length

```
str.length
```

don't put braces in this at last

Function → A standalone block of code that can be called directly.

Method → A function that belongs to an object or class.

String Indices

```
str[0], str[1], str[2]
```

some in-built fns of strings

can access strings each letter by this

Template Literals in JS

A way to have embedded expressions in strings

template literals are nothing but a special type of string in which we use ` backtick instead of " and has special properties

```
let temp = `this is a template literal`
```

String Interpolation

means the substitution of placeholders

To create strings by doing **substitution of placeholders**

```
`string text ${expression} string text`
```

can write expressions too
(meaning 1+2 or arithmetic
expressions)

in order to add a variables value in a string and u dont wanna do
code line number 100 u can simply just use a special type of string
called template literal in which u can do \${name} or \${obj.key} see
line number 103 for that

String Methods in JS

These are built-in functions to manipulate a string

str ke jagya string ka naam

- `str.toUpperCase( )`

like `name.toUpperCase();`

- `str.toLowerCase( )`

also these method do not change original value, they provide new updates value ,
so let `name="prince";` `name..toUpperCase();`
`console.log(name);` wont work coz original value would be the same.

- `str.trim( )` // **removes whitespaces**

present at the start and end of the string



String Methods in JS

- `str.slice(start, end?)` // returns part of string

abcde f g
↑ start ↑ end
abcde
0 1 2 3 4
└───┘

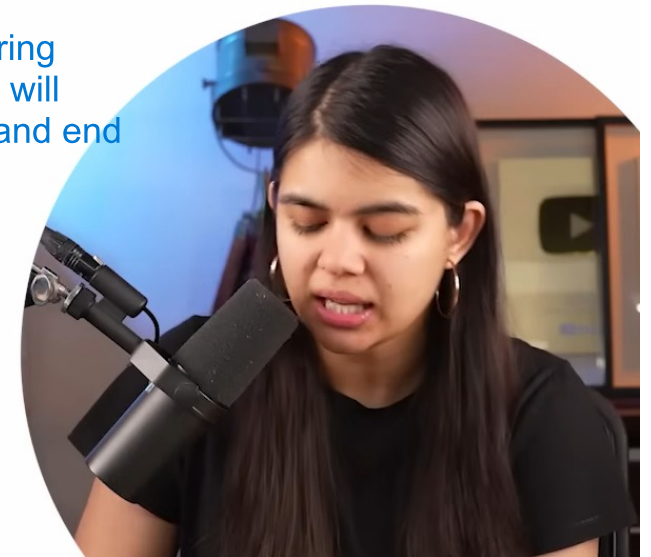
slice(start, end)
↓ ↓
1 3
X

- `str1.concat( str2 )` // joins str2 with str1

- `str.replace( searchVal, newVal )`

- `str.charAt( idx )`

in str slice u actually trim the string
but the end index which u write will
not be included oly start index and end
index ke pehle wala



String Methods in JS

- **str.slice(start, end?)** // returns part of string

- **str1.concat(str2)** // joins str2 with str1

- **str.replace(searchVal, newVal)**

concatnation can be done by two ways, one by method
and one by doing str1 +str2; this works too

- **str.charAt(idx)**

└
to know the character at some
specific index see line 145

used to replace some index ,by first seaching em and then replacing
em searchval should be the actual string letter , and newval too
dont write index numbers console.log(str.replace("h","y");

Let's Practice

Qs1. Prompt the user to enter their full name. Generate a username for them based on the input. Start username with @, followed by their full name and ending with the fullname length.

eg: user name = “shradhakhapra” , username should be “@shradhakhapra13”

```
let name= prompt("enter your name");  
let at="@";  
let length=name.length;  
let nEwStr=at.concat(name);  
console.log(nEwStr.concat(length));
```